

# Four Months' Notice

## Turning El Niño into a dengue early-warning alarm for ASEAN

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A freely available climate index gives ASEAN health offices months of warning before a dengue season turns severe. We turn that lag into a two-condition alert rule any provincial health office can run today — no model, no code, no paid data.

SDG 3 Health

SDG 13 Climate

SDG 6 Water

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Team [NAMA TIM] | [NAMA INSTITUSI] | Indonesia

## THE PROBLEM

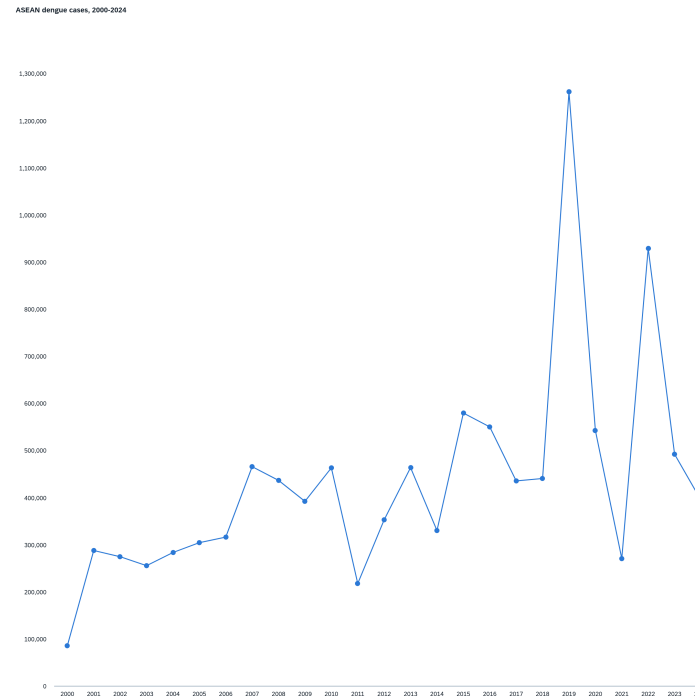
# Dengue is ASEAN's fastest-moving climate-sensitive disease

# 1.26M

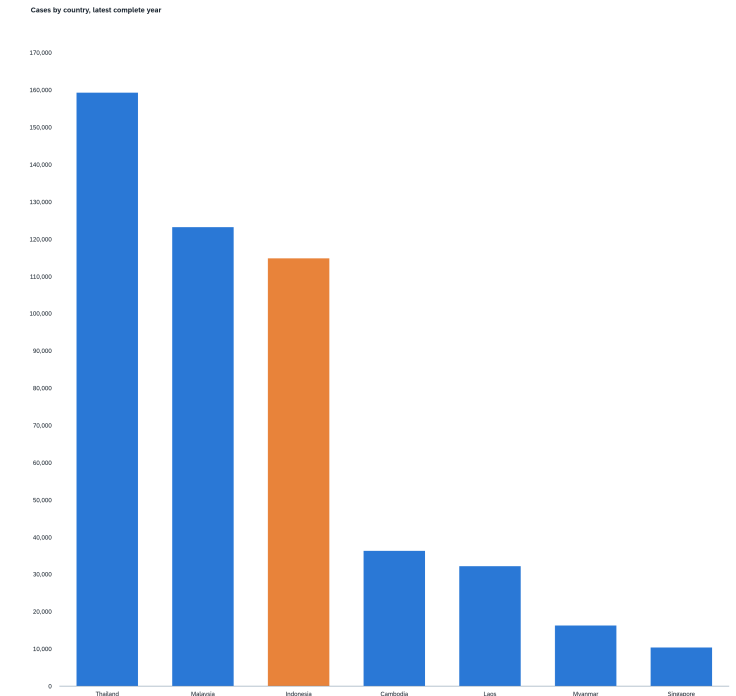
cases in 2019 — the worst year in our nine-country panel

- Vietnam 368k · Philippines 220k · Thailand 159k (latest complete year per country)
- The 2023–24 El Niño added an estimated **9.6 million cases worldwide** (Tian et al., 2025)
- Outbreaks are still fought **reactively** — after cases surge

### ASEAN dengue cases, 2000–2024



### Cases by country, latest complete year



## THE QUESTION

# Climate data arrives months before patients do

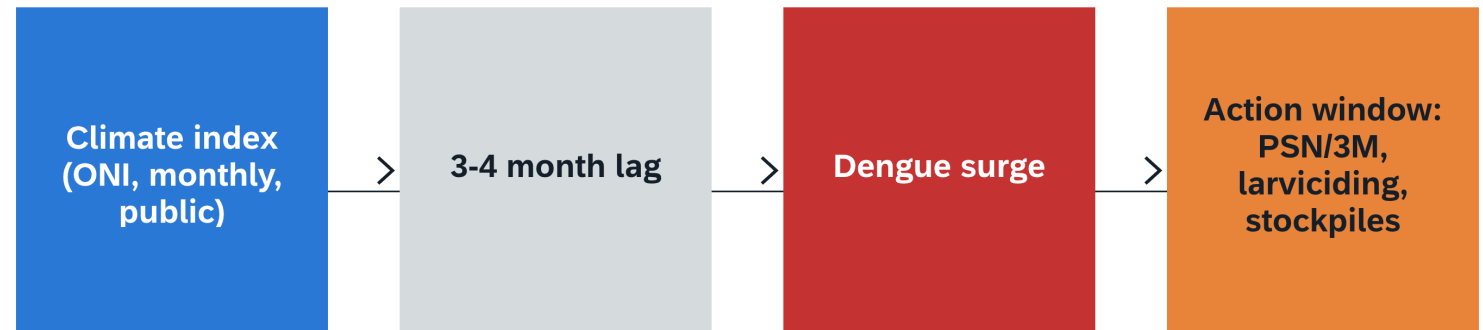
Can a freely available climate index tell a health office how bad the coming season will be — early enough to act?

- **SDG target 3.3** — end the epidemics of neglected tropical diseases; dengue is one of them
- **SDG target 13.1** — resilience and adaptive capacity to climate-related hazards
- **SDG target 6.1** — safe water for all; without piped water households store it, and *Aedes* breeds in the store

**ASEAN priority.** Dengue sits under **ASEAN Health Cluster 2** — *Responding to All Hazards and Emerging Threats* — whose ASEAN Dengue Day statements ask member states for earlier, multi-sector action.

Our test: does the signal survive when we remove the seasonal cycle, hold out unseen years, and use the WHO outbreak definition?

## From public index to public action



## DATA &amp; METHOD

# Nine open datasets, one monthly panel, zero paid sources

| DATA                                | ROWS   | COVERAGE                             |
|-------------------------------------|--------|--------------------------------------|
| Dengue cases (OpenDengue V1.3)      | 70,397 | 10 countries, 1955–2025              |
| Dengue × climate monthly panel      | 55,329 | province-month, lags<br>pre-computed |
| Temperature & rainfall (CCKP, ERA5) | 9,000  | monthly, 1950–2024                   |
| ONI El Niño index (NOAA CPC)        | 918    | monthly, 1950–2026                   |
| Water & sanitation (WHO/UNICEF JMP) | 8,692  | 2000–2024                            |
| Population (BPS Indonesia)          | 140    | provinces, denominators              |
| Flood exposure (GHSL)               | 30,348 | 836 cities, 1975–2030                |

SDG 3

SDG 13

SDG 6

Every table joins on the same two keys — iso3 + year (plus period for the monthly dengue × climate panel). Nothing is entered by hand.

## METHOD

- Monthly national series, 2010+
- Lagged correlations, 0–4 months
- **Anomalies** — seasonal cycle removed before we believe any correlation
- A two-condition trigger rule
- Out-of-sample test on unseen years
- WHO/PAHO endemic-channel benchmark

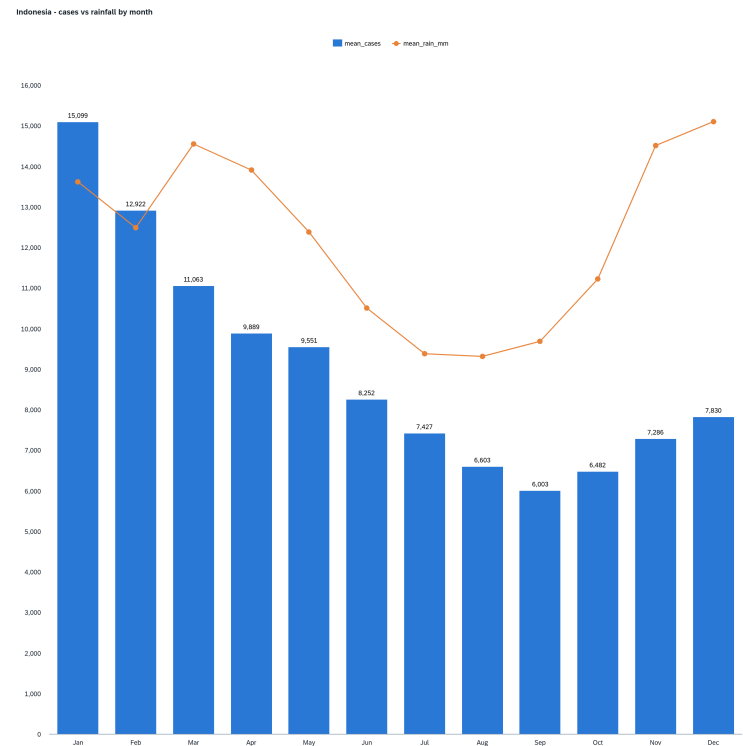
All charts built in SAP Analytics Cloud.

## ANALYSIS 1 — THE SEASON

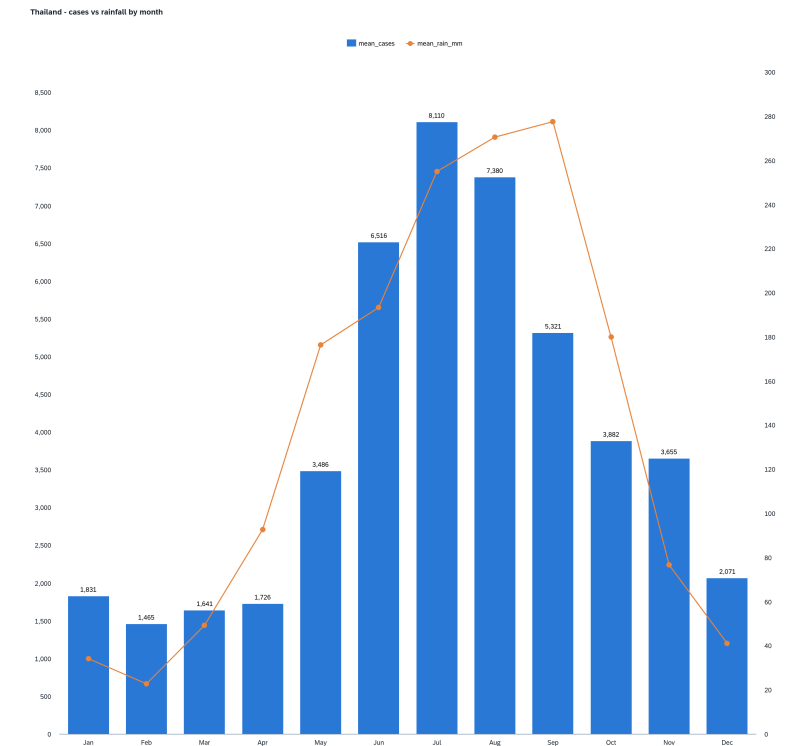
# Dengue has a calendar — and it is not the same calendar everywhere

- **Indonesia:** cases peak **Jan–Apr** (Jan avg 15,099/month vs Sep 6,003 —  $2.5\times$ ), right after the December rainfall peak
- **Thailand:** cases climb with the **onset** of the wet season and peak **Jun–Sep** (Jul avg 8,110 vs Feb 1,465 —  $5.5\times$ )
- Same disease, opposite halves of the year → a regional alarm must be **local in timing**

Indonesia — cases vs rainfall by month



Thailand — cases vs rainfall by month



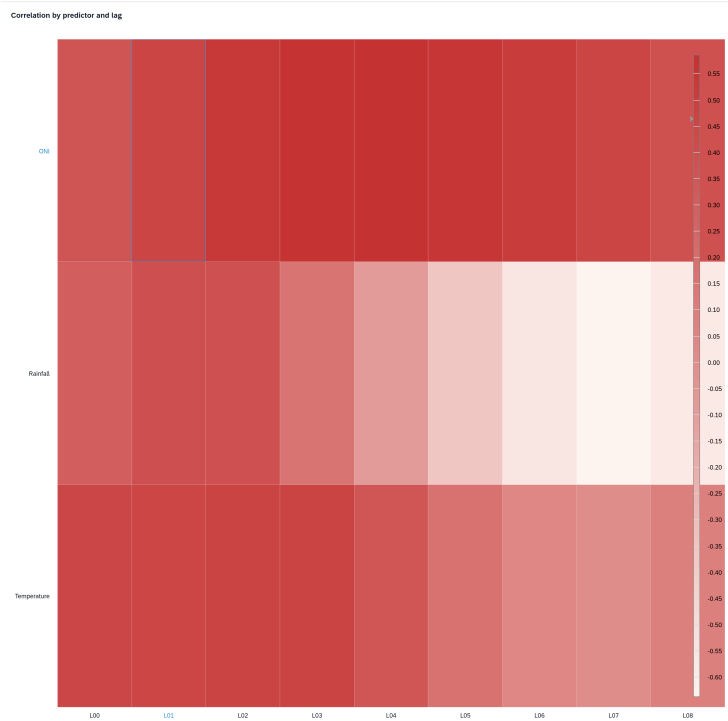
ANALYSIS 2 — THE LAG

The climate signal arrives 1–4 months before the cases

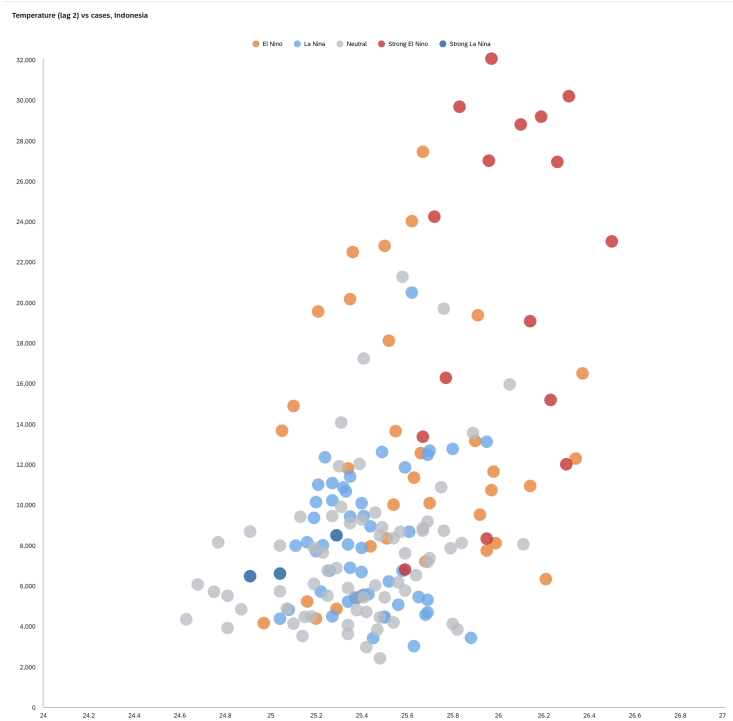
| COUNTRY   | BEST PREDICTOR | LAG     | R     |
|-----------|----------------|---------|-------|
| Indonesia | ONI (El Niño)  | 4 mo    | +0.59 |
| Indonesia | Temperature    | 3 mo    | +0.47 |
| Indonesia | Rainfall       | 1 mo    | +0.40 |
| Thailand  | Temperature    | 3 mo    | +0.57 |
| Thailand  | Rainfall       | same mo | +0.48 |

The lag *is* the opportunity: it is the time a health office has to act.

Correlation by predictor and lag



Temperature (lag 2) vs cases, Indonesia



## ANALYSIS 3 — THE ACE CARD

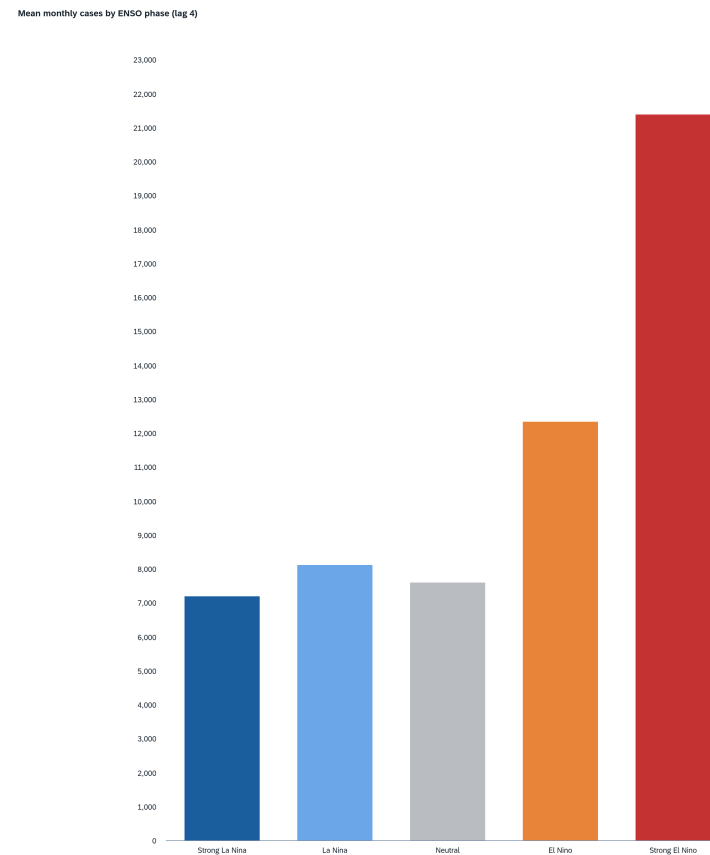
# A strong El Niño nearly triples Indonesia's monthly dengue burden

## 2.8×

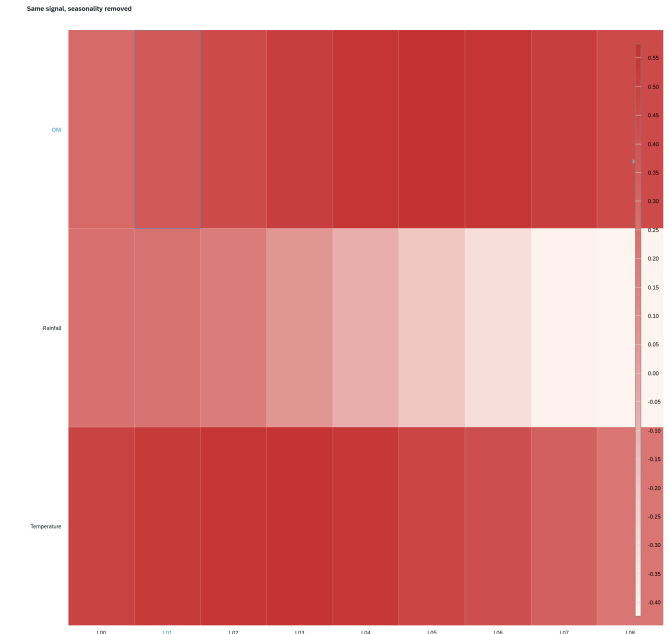
21,383 vs 7,597 cases per month, grouped by the ENSO phase four months earlier

- Grouped by the phase **four months earlier** — so this is a forecast, not hindsight
- **Not just seasonality**: after removing the seasonal cycle the ONI correlation holds at  **$r = +0.56$**
- **Not an aggregation artefact**: cases are higher in El Niño months in **33 of 34 Indonesian provinces** (sign test  $p = 2 \times 10^{-9}$ ) and 70 of 77 Thai provinces
- Rainfall does *not* survive the same test ( $+0.40 \rightarrow +0.24$ ) — we report that honestly

Mean monthly cases by ENSO phase (lag 4)



Same signal, seasonality removed

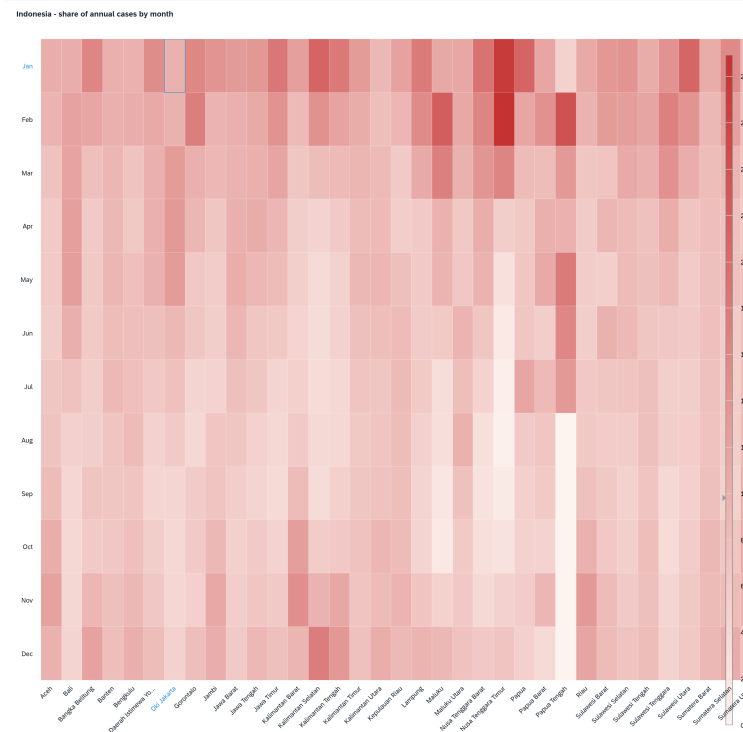


## ANALYSIS 4 — WHERE AND WHEN

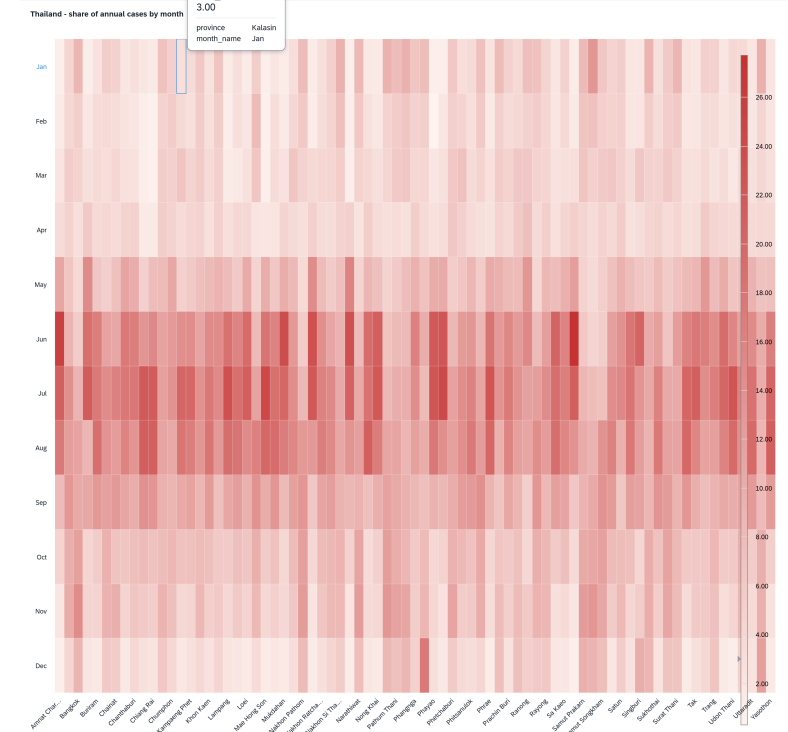
# The alarm must be set province by province

- **Indonesia:** 20 of 35 provinces peak in **January**, 8 in February — but **Bali peaks in May** and North Kalimantan in December
- **Thailand:** 38 of 77 provinces peak in **July**, 16 in August
- A single national alert date would be wrong for a third of Indonesia

Indonesia — share of annual cases by month



Thailand — share of annual cases by month

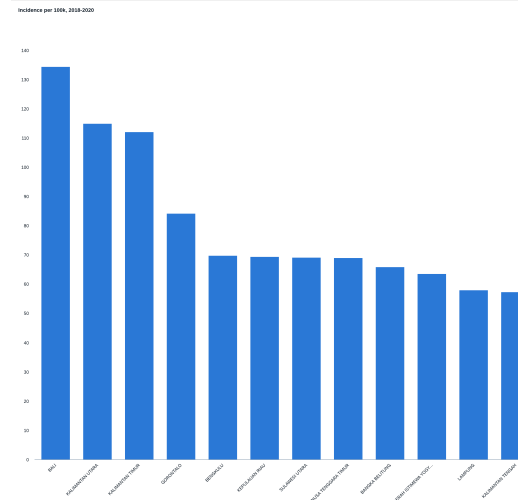


## ANALYSIS 5 — WHO GETS HIT HARDEST

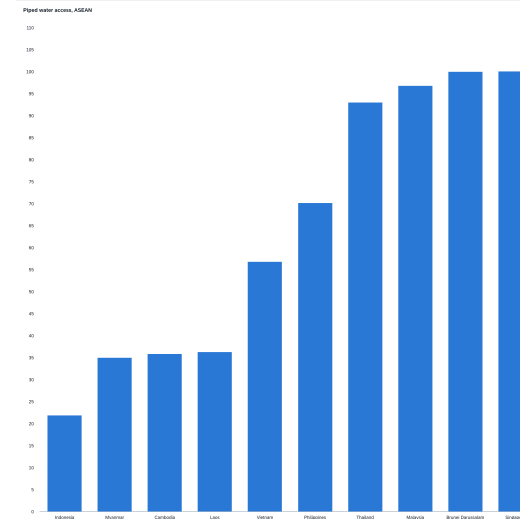
# The same alarm, aimed at the most exposed places

- Highest incidence 2018–2020: **Bali 134**, North Kalimantan 115, East Kalimantan 112 per 100k — **14× the lowest province** (Papua, 9)
- Indonesia has the **lowest piped-water access in ASEAN: 21.8%** (JMP 2024) — households store water, and stored water breeds *Aedes*
- **12.9 million people in Bangkok** and 1.7 million in Jakarta live in the 1-in-100-year flood zone
- We use this to **target** the alarm, not to explain it (see p. 14)

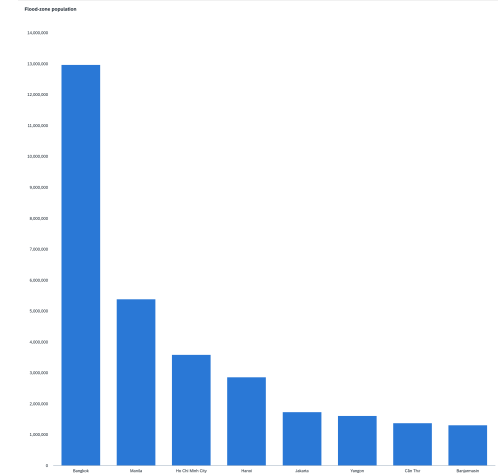
Incidence per 100k, 2018–2020



Piped water access, ASEAN



Flood-zone population

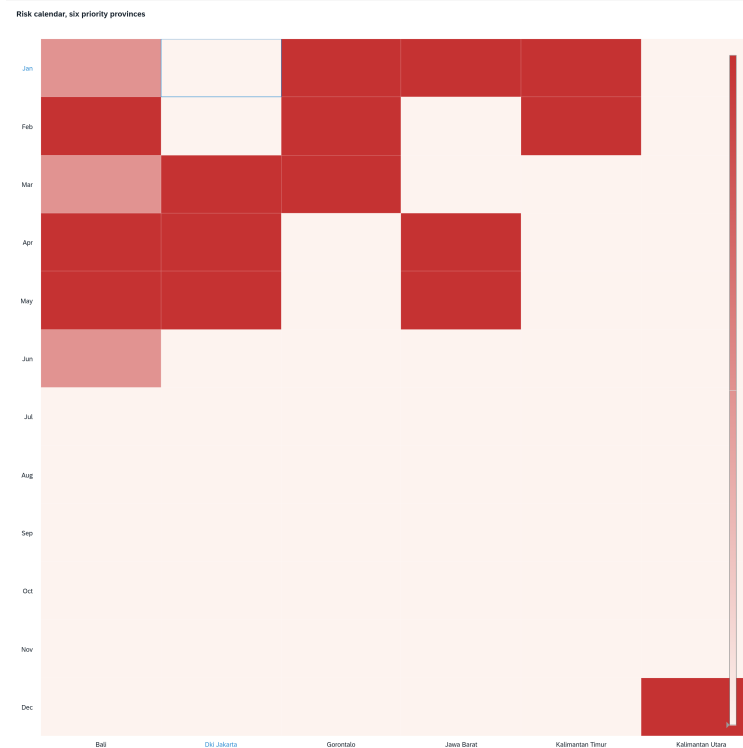


## SYNTHESIS

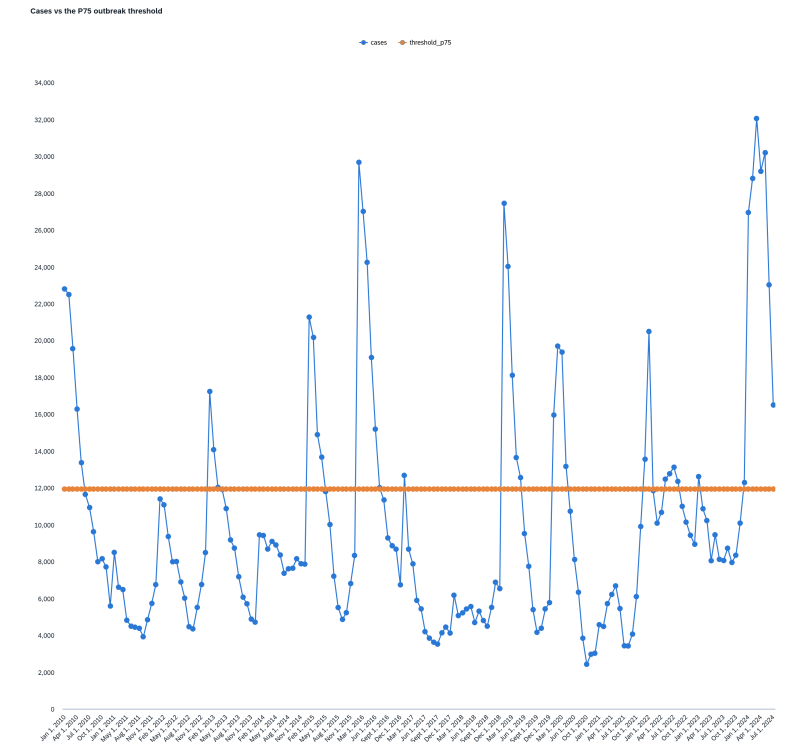
# From signal to schedule: a risk calendar every health office can read

- Each province gets three states — **Peak** · **Watch** · **Normal** — from its own case history
- The El Niño index sets **how severe** the coming peak will be; the calendar sets **when** it arrives
- Real lead times, every El Niño episode since 2009: **0–3 months** on observed ONI — and longer on NOAA/BMKG *forecast* ONI

Risk calendar, six priority provinces



Cases vs the P75 outbreak threshold



## THE SOLUTION

## Two conditions. No model. No code.

“If the month is Jan–Apr AND ONI four months ago was  $\geq 0.5 \rightarrow$  raise alert”

11.6%

of months fire the alert

100%

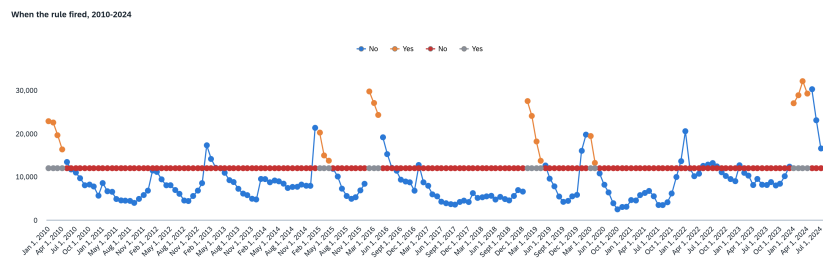
precision — every alert was a top-quartile dengue month

4×

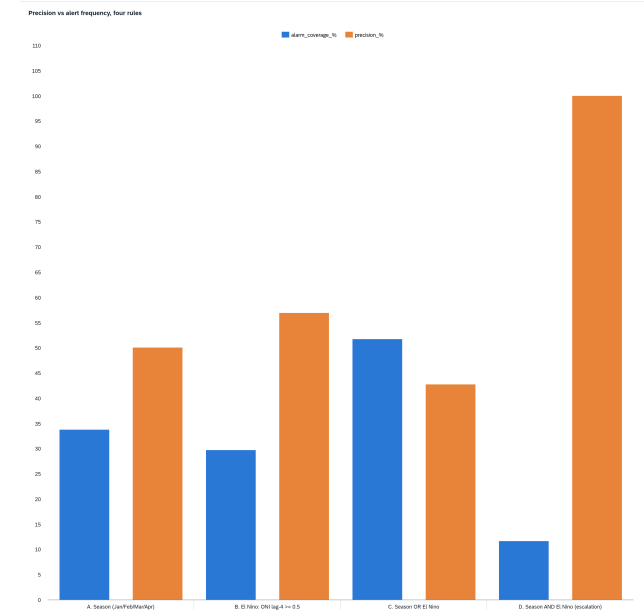
lift over the base rate

- Both inputs are public and free: NOAA publishes ONI monthly, the calendar comes from the office's own records
- Runs in a spreadsheet, a WhatsApp broadcast, or this SAC dashboard — competitors promise machine learning; we promise something a provincial health office can start next week

## When the rule fired, 2010–2024



## Precision vs alert frequency, four rules

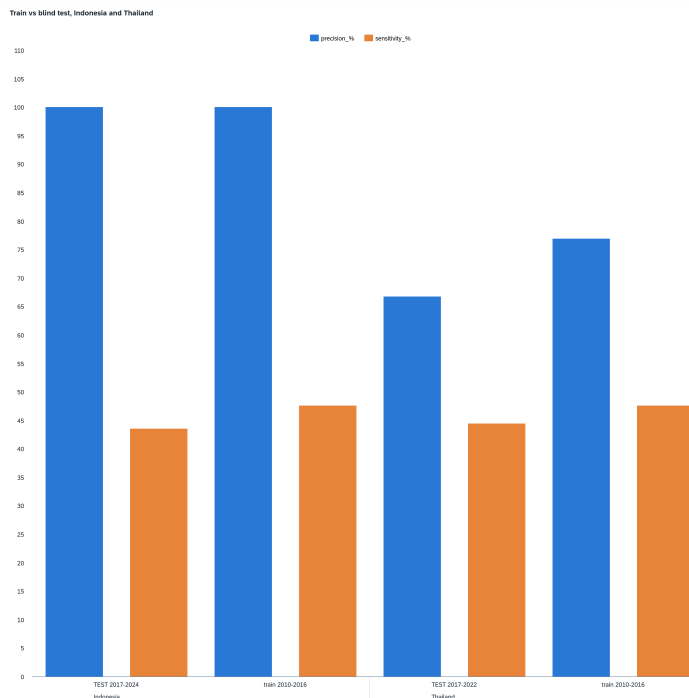


## VIABILITY

# We tried to break our own rule. It held.

- **Out-of-sample test:** rule built on 2010–2016, tested blind on 2017–2024 → **precision still 100%**, sensitivity 44%, lift 3.9
- Against the **WHO/PAHO endemic-channel** definition — which removes seasonality by construction — specificity **91%**, lift 2.5, and *all* of the value comes from the El Niño condition, not the season
- Published dengue EWS report PPV 43–86% (17-system review); we reach a comparable PPV **with no statistical model and monthly data**
- Replicates in Thailand with its own best predictor (temperature anomaly, lag 2): lift 2.7

## Train vs blind test, Indonesia and Thailand



## How we compare to published EWS

### How we compare to published EWS

[system](#) | [1](#)

| Measures                   |                         | sensitivity_low | sensitivity_high | precision_low | precision_high |
|----------------------------|-------------------------|-----------------|------------------|---------------|----------------|
| system                     | source                  |                 |                  |               |                |
| 17 published EWS (range)   | Baharom et al. 2022     | 50              | 100              | 43            | 86             |
| EWARS-csd, Colombia        | Schlesinger et al. 2024 | 97              | 97               | 75            | 75             |
| Rule D vs P75 (blind test) | this notebook           | 44              | 44               | 100           | 100            |
| Rule D vs endemic channel  | this notebook           | 29              | 29               | 30            | 30             |
| WHO-TDR EWARS, Mexico      | Cardenas et al. 2021    | 100             | 100              | 83            | 83             |

### ADOPTION IN THREE PHASES

- 1 Bali, N. & E. Kalimantan pilot →
- 2 all 38 provinces →
- 3 ASEAN peers

Cost: open data, monthly refresh, existing staff.

IMPACT — INDONESIA ALONE

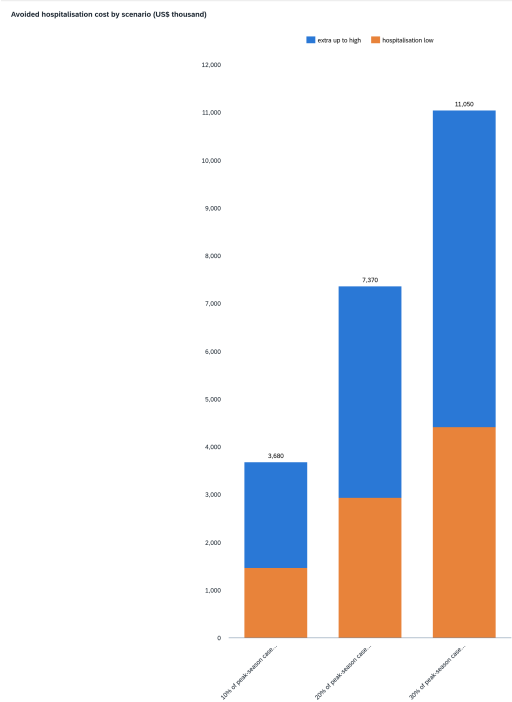
Acting 4–8 weeks earlier is worth US\$1.5–3.7 million a year

4,656

cases/year prevented in Indonesia, under the most conservative scenario

- Indonesia averages **104,753 cases/year** (complete years 2015–2023); **44% fall in Jan–Apr**
- Prevent just **10%** of peak-season cases (an assumption, not a finding) = 4,656 cases/year
- At Indonesia’s published hospitalisation cost of US\$316–791/case → **US\$1.5–3.7 M (Rp 24–59 billion)/year**
- For scale: dengue costs Indonesia **US\$381–681 M/year** (Nadjib 2019; Wilastonegoro 2020)
- Lead time is real: in the 2023–24 episode the alarm fired **Sep 2023**; cases crossed the peak threshold **Dec 2023**

Avoided hospitalisation cost by scenario



Lead time, every El Niño episode since 2009

Lead time, every El Nino episode since 2009

episode 1

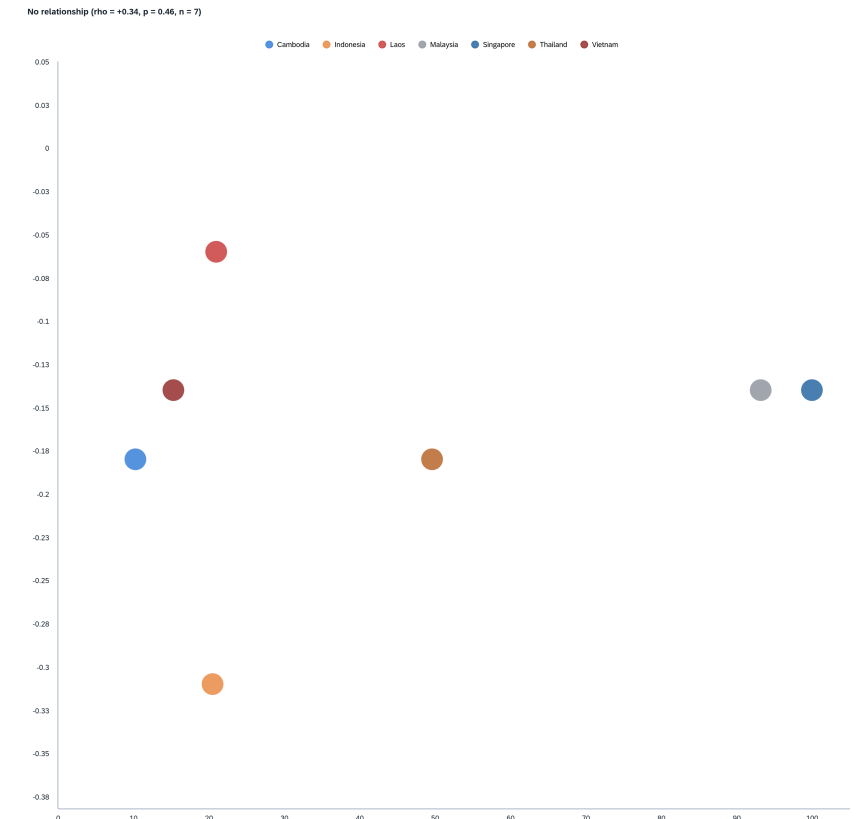
| episode | alarm              | cases_cross        | lead_time_months |
|---------|--------------------|--------------------|------------------|
| 2009/10 | Dec 1, 2009 (2009) | Jan 1, 2010 (2010) | 1                |
| 2014/16 | Feb 1, 2015 (2015) | Feb 1, 2015 (2015) | 0                |
| 2018/19 | Jan 1, 2019 (2019) | Jan 1, 2019 (2019) | 0                |
| 2019/20 | Mar 1, 2020 (2020) | Mar 1, 2020 (2020) | 0                |
| 2023/24 | Sep 1, 2023 (2023) | Dec 1, 2023 (2023) | 3                |

## LIMITS &amp; DATA ETHICS

# What we tested and threw away

- **We falsified our own innovation angle.** “Drought  $\times$  water access” looked promising ( $r = -0.31$ ) but collapsed after controlling for El Niño (partial  $r = -0.15$ ) and showed no moderation by piped access ( $\rho = +0.34$ ,  $p = 0.46$ ,  $n = 7$ ). **We dropped the claim.**
- **We do not claim a 2023–24 record** — our panel is incomplete for Vietnam and the Philippines after 2022.
- **Incidence only for 2018–2020**, the only years with official BPS denominators. Elsewhere we use raw counts and never rank regions by them.
- **Flood and disaster records look correlated with dengue ( $r = +0.66$ ) — until you divide by population ( $r = -0.14$ ).** We report the null.
- Correlation is not causation: this is an **early-warning** claim, with mechanism cited from the literature.

## No relationship ( $\rho = +0.34$ , $p = 0.46$ , $n = 7$ )



THE PRODUCT

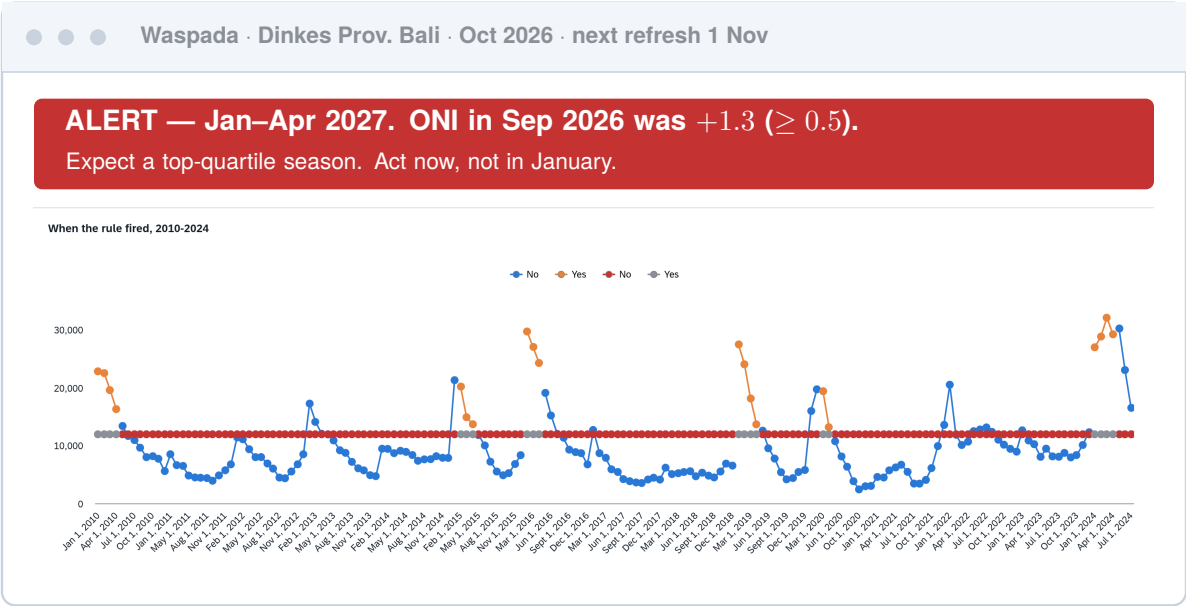
# Waspada — the rule, delivered on the first of every month

## Waspada

*waspada* — Indonesian and Malay for *stay alert*, and the first rung of Indonesia’s own hazard ladder. One SAC page a district officer opens on the 1st, plus an automatic WhatsApp advisory. It escalates; it never declares an emergency.

The digital layer does the fetching, not the deciding. SAC refreshes the NOAA ONI feed and the office’s own case calendar on a schedule, and SAC Smart Predict runs beside the rule as a second opinion. The alert itself stays the two-condition rule — auditable by hand, so nobody has to trust a model they cannot inspect.

### What Bali opens on 1 October 2026



### Who presses which button

| WHO                             | ROLE                                                                                                     |
|---------------------------------|----------------------------------------------------------------------------------------------------------|
| District health office gov't    | Reads the advisory; moves PSN/3M, larviciding and stockpiles forward into the alerted window             |
| Ministry of Health · BMKG gov't | Own the monthly refresh; BMKG’s forecast ONI stretches the notice past four months                       |
| Jumantik cadres NGO & community | Larval checks and household messaging concentrated in the alerted months instead of spread thin all year |
| Cloud & telco partners private  | Host the SAC tenant and the broadcast — one analyst-hour a month, no data team                           |

### SCALES BY SWAPPING ONE INPUT

Thailand already validates with its own predictor (temperature anomaly, lag 2, lift 2.7). Any ASEAN member repeats the fit on its own records.

## REFERENCES

# Data sources and literature

## DATA

OpenDengue V1.3 (doi:10.6084/m9.figshare.24259573.v4) · World Bank Climate Change Knowledge Portal (ERA5 0.25°) · NOAA CPC Oceanic Niño Index · WHO/UNICEF Joint Monitoring Programme 2025 · BPS Indonesia (WebAPI) · Global Human Settlement Layer (JRC) · SIPSN KLHK · BNPB.

## LITERATURE

Tian et al. 2025 *Nat Commun* · Djaafara et al. 2026 *PLOS NTD* · Sutriyawan et al. 2025 · Lowe et al. 2021 *Lancet Planet Health* · Gibb et al. 2023 *Nat Commun* · Baharom et al. 2022 · Cárdenas et al. 2021 · Schlesinger et al. 2024 · Hussain-Alkhateeb et al. 2018 (WHO-TDR EWARS) · Nadjib et al. 2019 *PLoS NTD* · Wilastonegoro et al. 2020 *AJTMH* · Sasmono et al. 2026 *IJID Regions* · Childs et al. 2025 *PNAS*.

**TODO sebelum submit:** ganti seluruh pranala consensus.app di `analisis/keluaran/referensi_per_klaim.md` dengan DOI jurnal, lalu lengkapi sitasi di halaman ini menjadi format penuh (penulis, tahun, judul, jurnal, DOI). Halaman referensi tidak dihitung dalam batas 15 halaman — boleh lebih dari satu halaman.